

Tau Analysis Project - Point of Situation

Project: Tau-Tau Analysis with Central Exclusive Production **Framework:** CMSSW 12.2.4 **Dataset:** 2018 Ultra Legacy (UL) **Date:** December 18, 2025

Executive Summary

This analysis studies tau lepton pair production in association with forward protons using 2018 CMS data. The analysis employs a multi-phase approach to filter and reconstruct events, focusing on three decay channels: MuTau, ETau, and TauTau. The project uses ROOT/RDataFrame for event processing and implements proton tagging requirements for central exclusive production studies.

Analysis Channels

1. MuTau Channel (Primary Focus)

- **Phase 0:** Initial skimming with 1 tau + 1 muon selection
- **Phase 1:** Particle identification and kinematic cuts
- **Phase 2:** Proton selection and tagging requirements

2. ETau Channel

- Electron-tau final state
- Similar phase structure to MuTau

3. TauTau Channel

- Di-tau final state
 - Control region and signal region separation
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Data Processing Pipeline

Phase 0 - Initial Skimming

Objective: Reduce dataset size by selecting events with basic topology

Selection Criteria: - Exactly 1 tau candidate - Exactly 1 muon candidate (MuTau) or electron (ETau) - HLT trigger: HLT_IsoMu24 (MuTau) or HLT_Ele32_WPTight (ETau) - Luminosity filtering for data

Status: [COMPLETED] - Data events after Phase 0: 698,931,141 - Proton variables saved in output

Phase 1 - Event Selection

Objective: Apply stringent identification and kinematic requirements

Selection Criteria:

Variable	Cut Value	Description
tau_id1	> 63	DeepTau2017v2p1 vs Jet
tau_id2	> 7	DeepTau2017v2p1 vs Electron
tau_id3	> 1	DeepTau2017v2p1 vs Muon
tau_pt	> 100 GeV	Tau transverse momentum
muon_id	>= 3	Muon ID (Medium or better)
muon_pt	> 35 GeV	Muon transverse momentum
delta_R	> 0.4	Angular separation
eta	< 2.4	Detector acceptance
Charge signs	Opposite (signal) / Same (QCD)	Lepton charges

Event Yields After Phase 1:

Sample	Starting Events	Phase 0	Phase 1	Efficiency
Data	984,127,247	19,698	6,859	0.0007%
QCD	984,127,247	19,698	1,011	0.0001%
ttjets	304,895,029	42,500,241	26,649	0.009%
DY	195,510,810	41,388,654	1,704	0.0009%
Signal	299,979	N/A	13,893	4.6%

Status: [COMPLETED] for most samples - [WARNING] Note: New processing with proton variables yields significantly different statistics

Phase 2 - Proton Selection

Objective: Select events with forward protons in PPS (Precision Proton Spectrometer)

Proton Selection Strategy:

1. **MultiRP Requirement:** Only multiRP protons used (better reconstruction)
2. **Two-Arm Topology:** Require at least one proton in each arm (0 and 1)
3. **Highest xi Selection:** For multiple protons per arm, select largest xi value
 - $\xi = 1 - (p_{\text{final}}/p_{\text{initial}})$: fractional momentum loss
 - Larger xi corresponds to primary scattered proton
 - Reduces pile-up contamination

Output Variables: - selected_xi_arm0: Proton momentum loss in arm 0 -
selected_xi_arm1: Proton momentum loss in arm 1

Status: [IN PROGRESS]

Technical Implementation

Software Stack

- **Framework:** CMSSW 12.2.4
- **Language:** Python 3 with ROOT/RDataFrame
- **Storage:** EOS (/eos/user/m/mbianco/)
- **Authentication:** VOMS proxy (CMS VO)

Key Technologies

- ROOT RDataFrame for columnar analysis
- XRootD for remote file access
- Multithreading enabled for performance
- Resume capability with JSON checkpointing

File Structure

```
tau_analysis/  
|-- MuTau_channel/          # MuTau analysis scripts  
|-- ETau_channel/           # ETau analysis scripts  
|-- TauTau_Channel/         # TauTau analysis scripts  
|-- fase0_code/             # Phase 0 processing  
|-- fase1_code/             # Phase 1 + CMSSW dependencies  
|-- plots/                  # Plotting scripts  
|-- background/             # Background estimation  
+-- output_plots*/          # Generated plots
```

Current Analysis Focus

Proton Variable Handling

Challenge: Initial approach of adding pileup protons post-processing was incorrect.

Solution: Save proton-related variables directly from NanoAOD during Phase 0: - nProton_multiRP / nProton_singleRP - Proton_multiRP_xi (momentum loss) - Proton_multiRP_arm (detector arm: 0 or 1) - Proton_multiRP_t (momentum transfer squared) - Proton_multiRP_thetaX/Y (scattering angles) - Proton_multiRP_time / timeUnc - Proton_singleRP_xi

Current Processing

File: fase1_qcd.py - Processing QCD sample with Phase 1 cuts - Same-sign requirement ($\text{muon_charge} \times \text{tau_charge} > 0$) - Output: /eos/user/m/mbiancco/samples_2018_mutable/fase1_qcd_proton_vars/

Analysis Variables

Kinematic Variables

Variable	Description
sist_mass	Invariant mass of mu+tau system
sist_pt	Transverse momentum of system
sist_rap	Rapidity of system
acop	Acoplanarity: $\text{abs}(\Delta\phi) / \pi$
met_pt	Missing transverse energy
delta_r	Delta-R separation between leptons

Jet Variables

Variable	Description
jet_pt	Leading jet pT
jet_btag	DeepCSV b-tagging score
n_b_jet	Number of b-tagged jets (> 0.4506)

Weights

- MC samples: Generator weights + scale factors
 - Data samples: weight = 1.0
 - Scale factors applied: Pileup, Tau ID, Muon ID
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Generated Plots

Plot Categories

1. **No Proton Requirements**
 - Baseline kinematic distributions
 - All Phase 1 selected events
2. **Proton Requirements (Data Only)**
 - Forward proton tagging applied
 - Two-arm requirement enforced
3. **Proton Requirements (Data + QCD)**

- QCD background with proton selection
- For background estimation validation

Key Distributions

- Invariant mass (`mass`)
- System rapidity (`rapidity_central`, `rapidity_matching`)
- Transverse momentum (`pt_central`, `tau_pt`)
- Mass difference (`mass_diff`)
- Acoplanarity (`aco`)
- Missing ET (`met`)

Known Issues and Concerns

1. Event Yield Discrepancy

Observation: Processing with saved proton variables shows different event counts

Sample Type	Original Phase 1	With Proton Vars	Ratio
Data	6,859	1,708	0.25
QCD	1,011	234	0.23

Hypothesis: Code changes for proton variable saving in Phase 1 may have introduced additional implicit cuts.

Action Required: - [DONE] Re-ran Phase 0 with correct script - [TODO] Investigating Phase 1 processing differences - [TODO] Validate proton variable preservation

2. Luminosity File

File: `dadosluminosidade.txt` referenced but usage unclear in current code - Lumi filtering lines are commented out in `fase1_qcd.py`

3. Background Estimation

QCD Background: Same-sign to opposite-sign extrapolation method - Phase 1 QCD processing uses same-sign requirement - Need transfer factor calculation for opposite-sign region

Next Steps

Immediate Actions

1. [DONE] Complete Phase 1 processing for QCD with proton variables
2. [TODO] Run Phase 1 for Data (opposite-sign) with proton variables
3. [TODO] Debug event count discrepancy between processing versions
4. [TODO] Implement Phase 2 proton selection on Phase 1 output

Short-Term Goals

- Validate proton variable integrity across all phases
- Generate comparison plots: with/without proton requirements
- Calculate QCD normalization factors
- Implement systematic uncertainties

Long-Term Objectives

- Statistical analysis and limit setting
 - Signal vs. background discrimination
 - Cross-section measurements
 - Publication preparation
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Data Samples

Monte Carlo Samples

- **DY** (Drell-Yan): $Z/\gamma^* \rightarrow \tau\tau$ background
- **ttjets**: Top quark pair production background
- **QCD**: Multijet background (data-driven)
- **Signal**: Central exclusive production of tau pairs (SM)

Data Sample

- **2018 UL**: CMS 2018 Ultra Legacy dataset
- **Trigger**: Single muon (IsoMu24) for MuTau channel
- **Integrated Luminosity**: $\sim 59.83 \text{ fb}^{-1}$ (2018)

Storage Locations

- Input: `/eos/home-m/mbiancco/samples_2018_mutau/`
 - Phase 0 output: `/eos/user/m/mbiancco/samples_2018_mutau/fase0_mutau_proton_vars/`
 - Phase 1 output: `/eos/user/m/mbiancco/samples_2018_mutau/fase1_qcd_proton_vars/`
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Collaboration and Resources

Dependencies

- TauPOG corrections and scale factors
- Pileup reweighting profiles
- B-tagging scale factors (DeepCSV medium WP: 0.4506)
- Tau ID scale factors (DeepTau2017v2p1)

Computing Resources

- EOS storage allocation
 - CERN lxplus/batch system access
 - VOMS proxy authentication
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Revision History

Date	Milestone	Notes
2025-12	Phase 0 complete	Re-ran with correct proton variable saving
2025-12	Phase 1 QCD	Processing with same-sign requirement
Current	Phase 1 Data	TODO: Process opposite-sign data
Future	Phase 2	Proton selection implementation

Contact and Documentation

Project Directory: /Users/utilizador/cernbox/tau_analysis/
README: README.md **Main README PDF:** README.pdf

Appendix: Cut Flow Summary

MuTau Channel Cut Flow

Cut Stage	Data	QCD (SS)	ttjets	DY	Signal
NanoAOD	984M	984M	305M	196M	300k

Cut Stage	Data	QCD (SS)	ttjets	DY	Signal
Phase 0	698M*	698M*	42.5M	41.4M	-
Phase 0 (skim)	19,698	19,698	42.5M	41.4M	-
HLT + 1mu+1tau	19,698	19,698	42.5M	41.4M	-
Lepton ID	-	-	-	-	-
Kinematic cuts	-	-	-	-	-
Charge requirement	6,859	1,011	26,649	1,704	13,893
Final Phase 1	6,859	1,011	26,649	1,704	13,893

*Initial skim with proton variable saving

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